



Group 3 | 04/30/20205

NAME OF PROJECT:

Real-time Drone Detection using Open MV

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PROJECT BRIEF

Deep Learning - Mini Project

Agenda

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Introduction

Developing a real time drone detection system using Edge Impulse and Open MV module.

Problem: Unauthorized drones in sensitive areas pose significant threats to security and safety.

This project addresses the need for a real-time drone detection system using deep learning deployed on an OpenMV camera, combined with a PC-based dashboard for monitoring detections.

Expected Impact:

- Detecting Unauthorized Drones
- Dashboard - Shows status **Clear / Drone Detected**
- Record the timestamps (Date & Time)

Technical Requirements

Hardware Requirements

- Open MV Module - RT1062

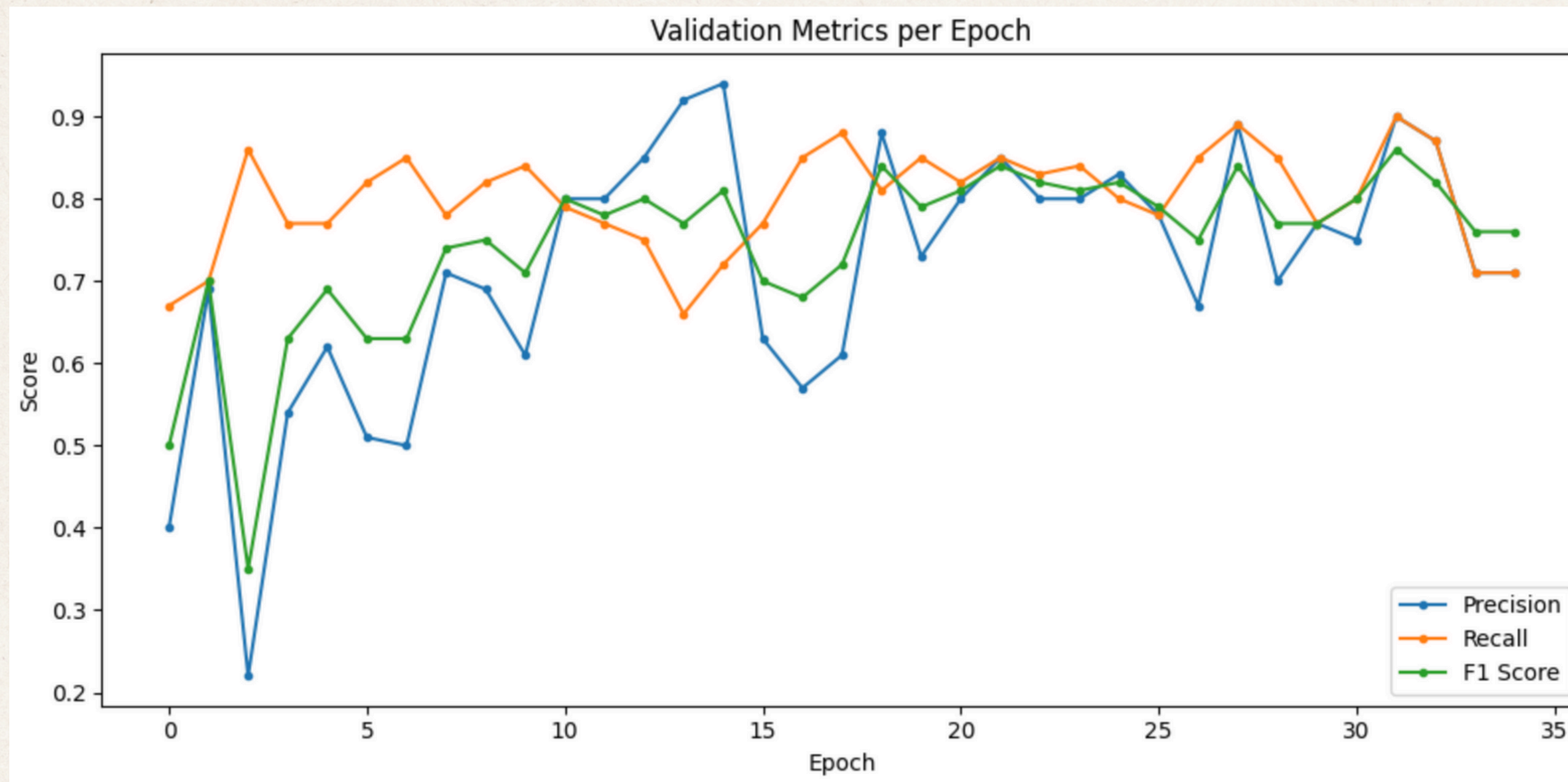
Software Requirements

- Python
- VS Code
- Gradio
- Edge Impulse
- Open MV IDE

Dataset and Training

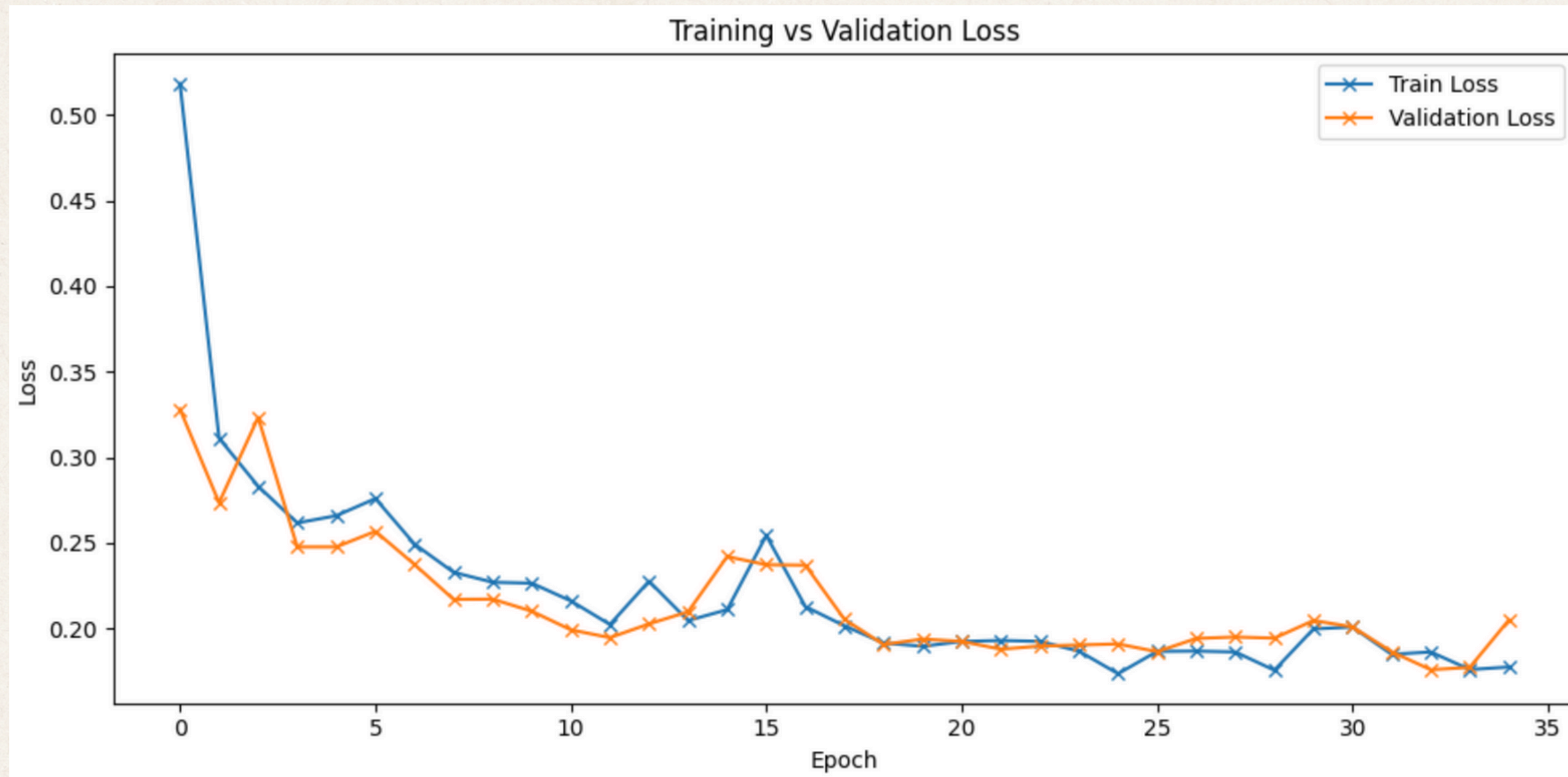
Used Two Datasets

1. Drone Dataset (Kaggle)
2. Yolo Drone Detection Dataset



Training Metrics - F1 recall vs Training Epochs

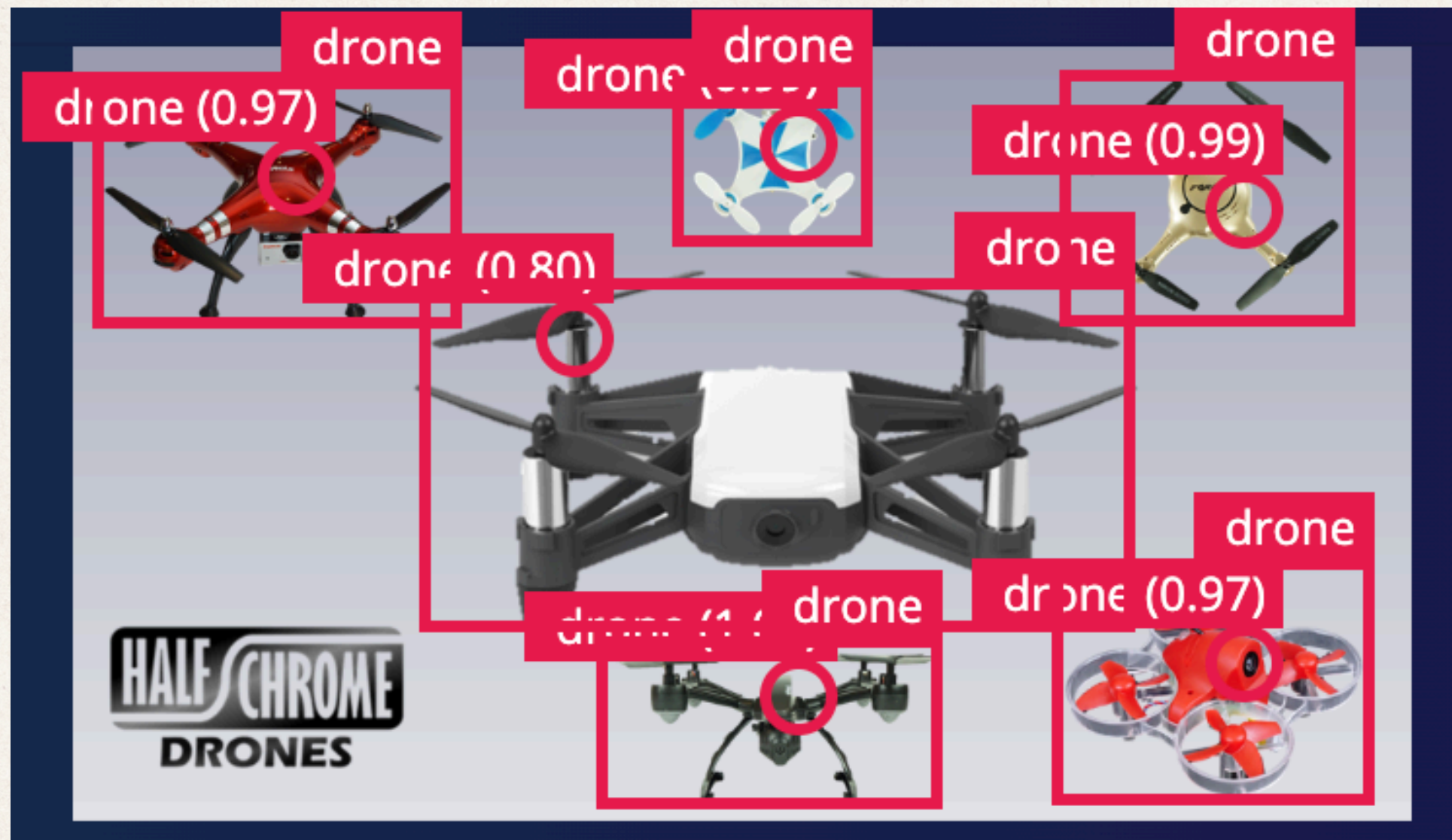
Dataset and Training



Training vs Validation Loss per epochs

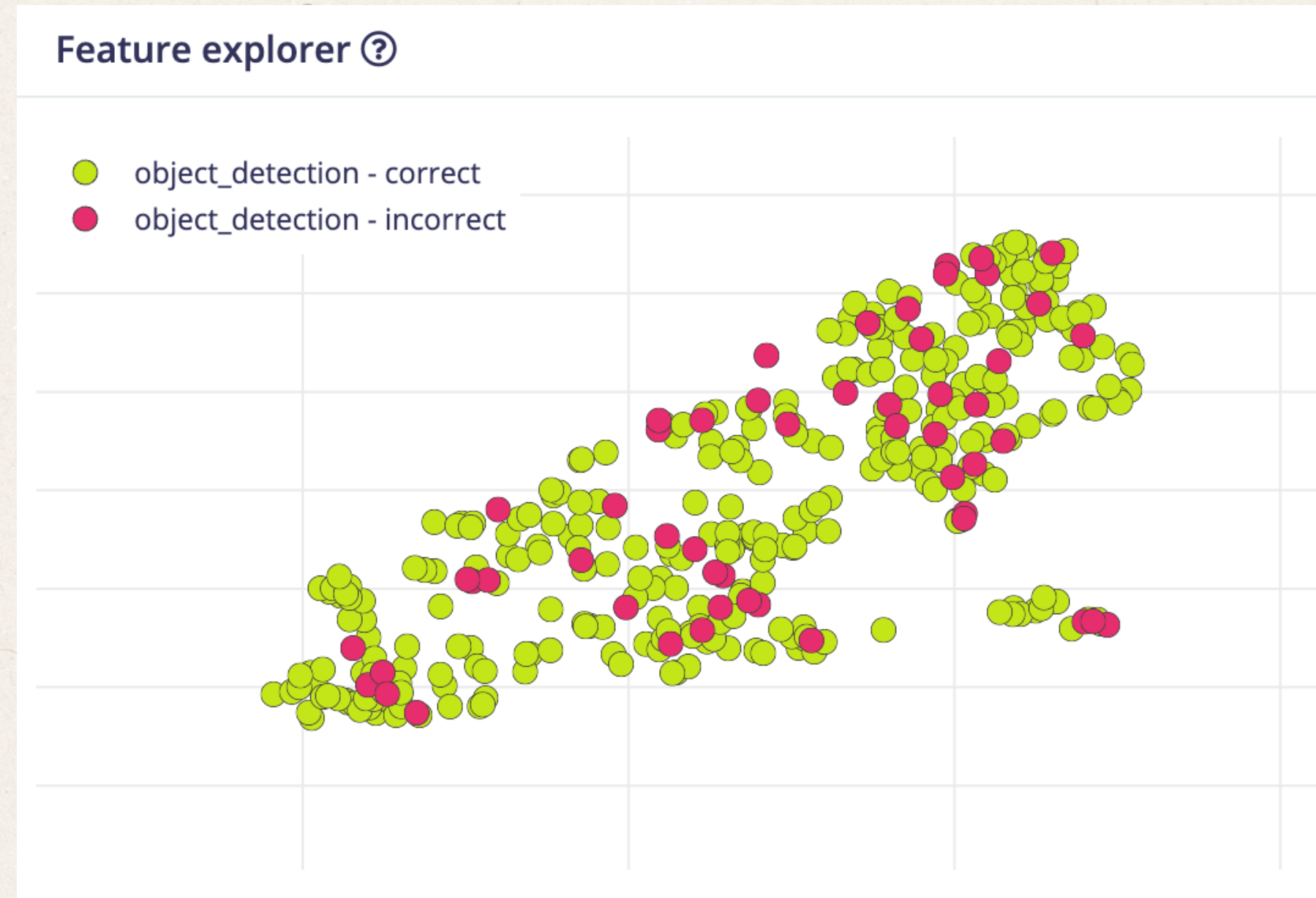
Results

- Test Accuracy: 86.02%
- Test Precision (non-background): 0.87
- Test Recall (non-background): 0.98
- Test F1 Score (non-background): 0.92



Results

08/11



Featurer Explorer - Correct vs. Incorrect Detections on Test Set

Challenges

1. Fine-tuning pretrained models:

- Resuming training on a TensorFlow Lite model exported from Edge Impulse while original FOMO MobileNetV2 model performed well.
- Handling model weight formats and optimizer compatibility in custom scripts proved non-trivial and time-consuming.

2. Local training complexities:

- Training object detection model locally using the exported Edge Impulse pipeline required replicating data augmentation, segmentation map logic and exact loss formulations.
- Partial success was achieved, inconsistencies in augmentation and label encoding occasionally impacted convergence.

3. OpenMV integration:

- Reading JPEG snapshots over UART from RT1062 and parsing detection coordinates in sync posed occasional synchronization issues.
- Required careful buffer handling in Python and OpenMV firmware to maintain reliable operation.

Future Scope

Multi-class detection:

Extend detection capabilities to differentiate between drone types or identify other aerial objects (eg birds, kites).

Audio Visual Fusion:

Integrate sound classification (eg drone propeller noise) alongside vision to reduce false positives and boost reliability.

GUI enhancements:

Add detection charts, snapshot saving, and alert sound notifications in the Tkinter UI to support real-world monitoring setups.

Integrate Alert Mechanisms:

Integrating alert mechanisms like Sound, SMS or Email notifications.

Thank you